

Question 2 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) Use the substitution $x = \ln u$ to find $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$ 3
- (b) Use one application of Newton's method to find an approximation to the root of the equation $\cos x = x$ near $x = 0.5$. Give your answer correct to two decimal places. 3
- (c) The curves $y = e^{2x}$ and $y = 1 + 4x - x^2$ intersect at the point $(0, 1)$. Find the angle between the two curves at this point of intersection. 3
- (d) (i) In how many ways can a committee of 2 Englishmen, 2 Frenchmen and 1 American be chosen from 6 Englishmen, 7 Frenchmen and 3 Americans. 2
- (ii) In how many of these ways do a particular Englishman and a particular Frenchman belong to the committee? 1

Question 1 (12 marks) Use a SEPARATE writing booklet

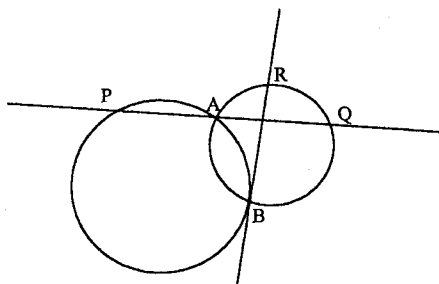
Marks

- (a) Solve $x(3-2x) > 0$ 2
- (b) Find $\frac{d}{dx} \{e^{-x} \cos^{-1} x\}$ 2
- (c) The remainder when $x^3 + ax^2 - 3x + 5$ is divided by $(x+2)$ is 11. Find the value of a . 2
- (d) Find the general solution of $2 \cos x + \sqrt{3} = 0$ 2
- (e) Solve $\frac{x^2 - 9}{x} \geq 0$ 2
- (f) Find $\int_0^2 (4+x^2)^{-1} dx$ 2

Question 4 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) Find the term independent of x in the expansion of $\left(3x - \frac{5}{x^3}\right)^8$ 3
- (b) Two circles cut at A and B. A line through A meets one circle at P and the other at Q. BR is a tangent to circle ABP and R lies on circle ABQ. Prove that $PB \parallel QR$. 3



- (c) The area bounded by the curve $y = \sin^{-1} x$ the y axis and the abscissa at $y = \frac{\pi}{2}$ is rotated about the y axis.

- (i) Show that the volume of the solid so formed is given

$$\text{by } \pi \int_0^{\frac{\pi}{2}} \sin^2 y \, dy$$

1

- (ii) Hence find the volume of this solid. 2

- (d) A particle moves in a straight line from a position of rest at a fixed origin O. its velocity is v when its displacement from O is x . 3

If its acceleration is $\frac{1}{(x+3)^3}$, find v in terms of x .

Question 3 (12 marks) Use a SEPARATE writing booklet

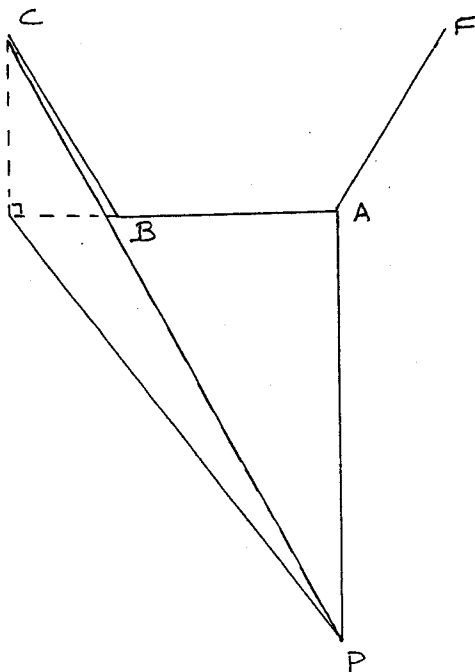
Marks

- (a) (i) Expand $\cos(\alpha + \beta)$ 1
- (ii) Show that $\cos 2\alpha = 1 - 2\sin^2 \alpha$ 1
- (iii) Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$ 1
- (b) If $\alpha = \tan^{-1}\left(\frac{5}{12}\right)$ and $\beta = \cos^{-1}\left(\frac{4}{5}\right)$, calculate the exact value of $\tan(\alpha - \beta)$. 2
- (c) A and B are the points $(-1, 7)$ and $(5, -2)$; P divides AB in the ratio $k : 1$.
- (i) Write down the coordinates of P in terms of k . 2
- (ii) If P lies on the line $5x - 4y = 1$, find the ratio of AP:PB 1
- (d) A biased coin has a probability of coming up heads equal to $p = 0.6$. Find the probability of getting exactly four heads in ten tosses of the coin. 1
- (e) Use mathematical induction to prove that 3
- $$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$$

Question 6 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) Solve $\cos x - \sqrt{3} \sin x = 1$, where $0 \leq x \leq 2\pi$ 3
- (b) Wheat falls from an auger onto a conical pile at the rate of $20\text{cm}^3\text{s}^{-1}$. The radius of the base of the pile is always equal to half its height.
- (i) Show that $V = \frac{1}{12}\pi h^3$ and hence find $\frac{dh}{dt}$ 2
- (ii) Find the rate at which the pile is rising when it is 8 cm deep 1
- (iii) Find the time taken for the pile to reach a height of 8 cm. 2
- (c) In a horizontal triangle APB, $AP=2AB$, and the angle A is a right angle. On AB stands a vertical and regular hexagon ABCDEF. Prove that PC is inclined to the horizontal at an angle whose tangent is $\frac{\sqrt{3}}{5}$. 4



Question 5 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) The speed $v \text{ m s}^{-1}$ of a particle moving along the X axis is given by $v^2 = 24 - 6x - 3x^2$, where $x \text{ m}$ is the distance of the particle from the origin.
- (i) Show that the particle is executing Simple Harmonic Motion 2
- (ii) Find the amplitude and the period of the motion 2
- (b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$
- (i) Show that the equation of PQ is given by the equation $y - \frac{1}{2}(p+q)x + apq = 0$. 2
- (ii) Find the condition that PQ passes through the point $(0, -a)$. 1
- (iii) If the focus of the parabola is S, prove that $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$ 2
- (c) One root of $x^3 + px^2 + qx + r = 0$ equals the sum of the two other roots, prove that $p^3 + 8r = 4pq$ 3

Question 7 (12 marks) Use a SEPARATE writing booklet

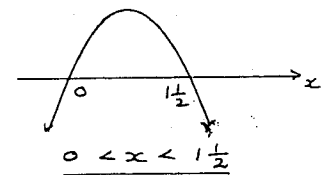
Marks

- (a) (i) Find the largest possible domain of positive values for which $f(x) = x^2 - 6x + 13$ has an inverse. 1
- (ii) Find the equation of the inverse function, $f^{-1}(x)$. 2
- (b) (i) Write down the expansions of $(1+x)^n$ 1
- (ii) Using part (i) show that 3
- $${}^{10}C_0 + {}^{10}C_2 7^2 + {}^{10}C_4 7^4 + {}^{10}C_6 7^6 + {}^{10}C_8 7^8 + {}^{10}C_{10} 7^{10} = 2^9 (2^{20} + 3^{10})$$
- (c) A projectile is fired from O, with velocity $V \text{ ms}^{-1}$, at an angle of α to the horizontal. After t seconds, its horizontal and vertical displacements from O are x metres and y metres.
- (i) Beginning with the acceleration equations $\ddot{x} = 0$ and $\ddot{y} = -g$ 2
- show that the equations of motion of the projectile are
- $$x = Vt \cos \alpha \text{ and } y = \frac{-gt^2}{2} + Vt \sin \alpha$$
- (ii) Fire fighters are forced to stay 60 metres away from a dangerous 3
- fire burning in a low open tank on horizontal ground. They have two pumps. One which can eject water in any direction at 30 ms^{-1} , is on the ground, while the other, which can eject water at 40 ms^{-1} but only horizontally, is on a vertical stand 5 m high.
- Take $g = 10 \text{ ms}^{-2}$.
- Using the equations of motion derived in (i) find whether or not both pumps can reach the fire.

End of Examination

Question 1

a) $x(3-2x) > 0$

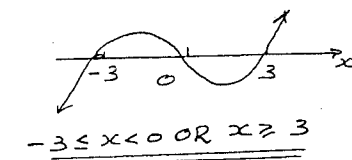


b) $\frac{d}{dx}(e^{-x} \cos^{-1} x)$
 $= e^{-x} \cdot \frac{-1}{\sqrt{1-x^2}} + (-e^{-x}) \cos^{-1} x$
 $= -e^{-x} \left(\frac{1}{\sqrt{1-x^2}} + \cos^{-1} x \right)$

c) $P(-2) = 11$
 $(-2)^3 + a(-2)^2 - 3(-2) + 5 = 11$
 $-8 + 4a + 6 + 5 = 11$
 $4a = 11$
 $a = \frac{11}{4}$

d) $2\cos x + \sqrt{3} = 0$
 $\cos x = -\frac{\sqrt{3}}{2}$
 $x = 2n\pi \pm \cos^{-1}(-\frac{\sqrt{3}}{2})$
 $= 2n\pi \pm (\pi - \cos^{-1}(\frac{\sqrt{3}}{2}))$
 $= 2n\pi \pm (\pi - \frac{\pi}{6})$
 $= 2n\pi \pm \frac{5\pi}{6}$

e) $\frac{x^2-9}{x} \geq 0$
 $x(x^2-9) \geq 0 \times x^2$
 $x(x^2-9) \geq 0$
 $x(x-3)(x+3) \geq 0$



f) $\int_0^2 \frac{1}{4+x^2} dx$
 $= \frac{1}{2} \left[\tan^{-1} \frac{x}{2} \right]_0^2$
 $= \frac{1}{2} \left\{ \tan^{-1} 1 - \tan^{-1} 0 \right\}$
 $= \frac{1}{2} \left(\frac{\pi}{4} - 0 \right)$
 $= \frac{\pi}{8}$

Question 2.

a) $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$ let $x = \ln u$
 $\frac{dx}{du} = \frac{1}{u}$
 $dx = \frac{1}{u} du$
 $u = e^x$
 $= \int \frac{u}{\sqrt{1-u^2}} \cdot \frac{1}{u} du$
 $= \int \frac{du}{\sqrt{1-u^2}}$
 $= \sin^{-1} u + C$
 $= \sin^{-1} e^x + C$

OR. let $x = \ln u$
 $\therefore u = e^x$
 $\frac{du}{dx} = e^x$
 $du = e^x dx$
 $\therefore \int \frac{e^x}{\sqrt{1-e^{2x}}} dx$
 $= \int \frac{du}{\sqrt{1-u^2}}$
 $= \sin^{-1} u + C$
 $= \sin^{-1} e^x + C$

b) $\cos x = x$
 $\therefore \cos x - x = 0$
Let $f(x) = \cos x - x$
 $f'(x) = -\sin x - 1$
Now $x_0 = 0.5$
 $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$x_1 = 0.5 - \frac{(\cos 0.5 - 0.5)}{(-\sin 0.5 - 1)}$
 $= 0.76$

c) $y = e^{2x}$
 $\frac{dy}{dx} = 2e^{2x}$
at $x=0$, $\frac{dy}{dx} = 2e^0 = 2$
 $y = 1 + 4x - x^2$
 $\frac{dy}{dx} = 4 - 2x$
at $x=0$, $\frac{dy}{dx} = 4$
 $\therefore m_1 = 4, m_2 = 2$
 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{4 - 2}{1 + 4 \times 2} \right|$
 $= \left| \frac{2}{9} \right|$
 $\theta = 12^\circ 32'$

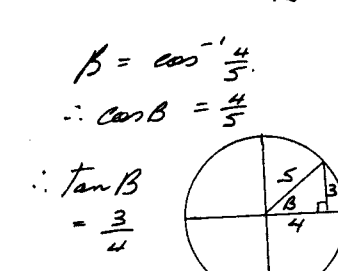
d) (i) $\frac{1}{2} \times \frac{7}{2} \times \frac{3}{1}$
 $= 15 \times 2 \times 3$
 $= 945$
(ii) $\frac{5}{1} \times \frac{6}{1} \times \frac{3}{1}$
 $= 90$

Question 3

(i) $\cos(\alpha + \beta)$
 $= \cos \alpha \cos \beta - \sin \alpha \sin \beta$
(ii) Let $\beta = \alpha$
 $\therefore \cos(\alpha + \alpha)$
 $= \cos \alpha \cos \alpha - \sin \alpha \sin \alpha$
 $= \cos^2 \alpha - \sin^2 \alpha$
 $= 1 - \sin^2 \alpha - \sin^2 \alpha$
 $= 1 - 2\sin^2 \alpha$
 $\therefore \cos 2\alpha = 1 - 2\sin^2 \alpha$

(iii) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$
 $= \lim_{x \rightarrow 0} \frac{1 - (1 - 2\sin^2 x)}{x^2}$
 $= \lim_{x \rightarrow 0} \frac{2\sin^2 x}{x^2}$
 $= 2 \cdot \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{\sin x}{x}$
 $= 2 \cdot 1 \cdot 1$
 $= 2$

b) $\tan(\alpha - \beta)$
 $= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$
Now $\alpha = \tan^{-1} \frac{5}{12}$
 $\therefore \tan \alpha = \frac{5}{12}$
 $\beta = \cos^{-1} \frac{4}{5}$
 $\therefore \cos \beta = \frac{4}{5}$



Hence $\tan(\alpha - \beta)$
 $= \frac{\frac{5}{12} - \frac{3}{4}}{1 + \frac{5}{12} \times \frac{3}{4}}$
 $= \frac{20 - 36}{48 + 15}$
 $= \frac{-16}{63}$

c) (i) $(-1, 7)$ and $(5, -2)$
 $k : 1$
 $x = 1 \times (-1) + k \times 5$
 $k + 1$
 $= -1 + 5k$
 $k + 1$
 $y = 1 \times 7 + k(-2)$
 $k + 1$
 $= 7 - 2k$
 $k + 1$

$\therefore P$ is $(\frac{-1+5k}{k+1}, \frac{7-2k}{k+1})$
(ii) P lies on $5x - 4y = 1$
 $\therefore 5(\frac{5k-1}{k+1}) - 4(\frac{7-2k}{k+1}) = 1$

$25k - 5 - 28 + 8k = k + 1$
 $33k - 33 = k + 1$
 $32k = 34$
 $k = \frac{34}{32}$
 $k = \frac{17}{16}$

$\therefore AP : PB = 17 : 16$

d) $P(x=4)$
 $= {}^{10}P_4$
 $= {}^{10}P_4$
 $= {}^{10}P_4 (0.4)^6 (0.6)^4$
 $= 0.1115$ To 4 dec. pl.

e) Step 1.
If $n=1$, L.H.S. = $1 \times 1!$
R.H.S. = $(1+1)! - 1$
 $= 2! - 1$
 $= 1$

$\therefore L.H.S. = R.H.S.$
Result is true for $n=1$.

Step 2
Assume that the result is true for $n=k$
i.e.
 $1 \times 1! + 2 \times 2! + \dots + k(k-1)! = (k+1)! - 1$

Step 3.
Prove that the result is true for $n=k+1$
i.e.
 $1 \times 1! + 2 \times 2! + \dots + (k+1)(k)! = (k+2)! - 1$

Proof:
L.H.S.
 $= 1 \times 1! + 2 \times 2! + \dots + (k+1)(k)! + (k+1)(k+1)!$
 $= (k+1)! - 1 + (k+1)(k+1)!$
 $= (k+1)! + (k+1)(k+1)! - 1$
 $= (k+1)!(1 + k+1) - 1$
 $= (k+1)!(k+2) - 1$
 $= (k+2)! - 1$
 $= R.H.S.$
Hence the result is true for $n=k+1$ if it is true for $n=k$

Step 4:

Since the result is true for $n=1$ then it is true for $n=1+1$ i.e. for $n=2$ and thus for $n=3$ and so on for all positive integral values of n .

Question 4

$$a) T_{k+1} = \binom{n}{k} a^{n-k} b^k$$

$$\therefore \text{for } (3x + \frac{-5}{x^3})^8$$

$$T_{k+1} = \binom{8}{k} (3x)^{8-k} (\frac{-5}{x^3})^k$$

$$= \binom{8}{k} 3^{8-k} x^{8-k} (-1)^k 5^k x^{-3k}$$

$$= \binom{8}{k} (-1)^k 3^{8-k} 5^k x^{8-4k}$$

$$= \binom{8}{k} (-1)^k 3^{8-k} 5^k x^{8-4k}$$

We want

$$8-4k=0$$

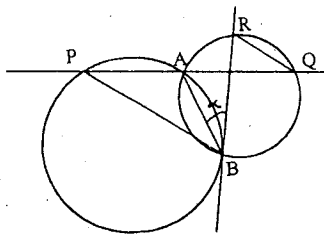
$$\therefore 4k=8$$

$$\therefore T_{2+1}$$

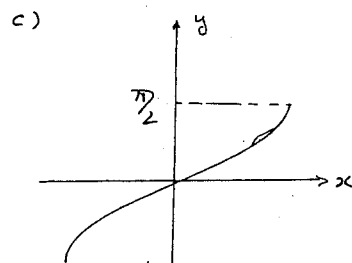
$$= \binom{8}{2} (-1)^2 3^{8-2} 5^2 x^0$$

$$= 28 \times 3^6 \times 5^2$$

$$= 510300$$



Join A to B and let $\angle ABR = \alpha$
 $\angle APB = \angle ABR$ (alt. ang. theorem)
 $\angle AQR = \angle ABR$ ($\angle$'s at the circumference standing on the same arc are equal)
 $\therefore \angle APB = \angle AQR$
 but there are alt. $\angle$'s
 $\therefore PB \parallel QR$



$$\text{Now } y = \sin^{-1} x$$

$$\therefore x = \sin y$$

$$x^2 = \sin^2 y$$

$$\therefore V = \pi \int_0^{\pi/2} x^2 dy$$

$$= \pi \int_0^{\pi/2} \sin^2 y dy$$

$$= \pi \int_0^{\pi/2} \frac{1}{2} (1 - \cos 2y) dy$$

$$= \frac{\pi}{2} \int_0^{\pi/2} (1 - \cos 2y) dy$$

$$= \frac{\pi}{2} \left[y - \frac{1}{2} \sin 2y \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} \left\{ \left(\frac{\pi}{2} - \frac{1}{2} \sin 2 \times \frac{\pi}{2} \right) - \left(0 - \frac{1}{2} \sin 0 \right) \right\}$$

$$= \frac{\pi}{2} \left\{ \frac{\pi}{2} - \frac{1}{2} \sin \pi - (0-0) \right\}$$

$$= \frac{\pi}{2} \left(\frac{\pi}{2} - 0 \right)$$

$$= \frac{\pi^2}{4} \text{ cubic units}$$

$$d) \ddot{x} = \frac{1}{(x+3)^2}$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{1}{(x+3)^2}$$

$$= (x+3)^{-2}$$

$$\therefore \frac{1}{2} v^2 = \int (x+3)^{-2} dx$$

$$\frac{1}{2} v^2 = \frac{(x+3)^{-1}}{-1 \times 1} + C$$

$$\frac{1}{2} v^2 = -\frac{1}{x+3} + C$$

$$\text{at } t=0, x=0, v=0$$

$$\therefore 0 = -\frac{1}{3} + C$$

$$C = \frac{1}{3}$$

$$\therefore \frac{1}{2} v^2 = -\frac{1}{x+3} + \frac{1}{3}$$

3.

$$\frac{1}{2} v^2 = -3 + x + 3$$

$$\frac{1}{2} v^2 = \frac{x}{3(x+3)}$$

$$\frac{1}{2} v^2 = \frac{x}{3(x+3)}$$

$$v^2 = \frac{2x}{3(x+3)}$$

$$v = \pm \sqrt{\frac{2x}{3(x+3)}}$$

as $a > 0$ for all real values of x then $v > 0$ for all real x

$$\therefore v = \sqrt{\frac{2x}{3(x+3)}}$$

Question 5.

$$a) v^2 = 24 - 6x - 3x^2$$

$$(i) \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$= \frac{d}{dx} \left\{ 12 - 3x - \frac{3}{2} x^2 \right\}$$

$$= -3 - 3x$$

$$= -3(x+1)$$

$$\text{let } y = x+1$$

$$\therefore \frac{dy}{dt} = \frac{dx}{dt} + \frac{dy}{dt} = \frac{dx}{dt}$$

$$\text{i.e. } \dot{y} = \dot{x}$$

$$\text{and } \ddot{y} = \ddot{x}$$

$$\therefore \ddot{y} = -3y \text{ which is S.H.M. centre } y=0$$

$$\text{i.e. } x+1=0$$

$$x=-1$$

$$(ii) \text{ Particle is at rest when } v=0$$

$$\text{i.e. } 24 - 6x - 3x^2 = 0$$

$$x^2 + 2x - 8 = 0$$

4.

$$x^2 + 2x - 8 = 0$$

$$(x+4)(x-2) = 0$$

$$x = -4 \text{ OR } x = 2$$

$$\therefore \text{length of path} = 2a = 6$$

$$\therefore a = 3$$

$$\text{Amplitude} = 3 \quad \text{Period} = \frac{2\pi}{\omega}$$

$$= \frac{2\pi}{\sqrt{3}}$$

b) (i)

$$m_{pq} = \frac{ap^2 - aq^2}{2ap - 2aq}$$

$$= \frac{a(p^2 - q^2)}{2a(p - q)}$$

$$= \frac{(p - q)(p + q)}{2(p - q)}$$

$$= \frac{p + q}{2}$$

Equation of PQ is

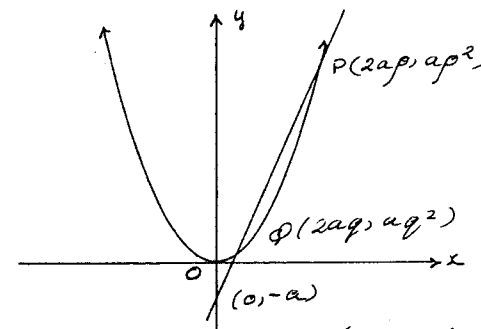
$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = \frac{1}{2}(p+q)(x - 2ap)$$

$$y - ap^2 = \frac{1}{2}(p+q)x - 2ap \times \frac{1}{2}(p+q)$$

$$y - ap^2 = \frac{1}{2}(p+q)x - ap^2 - aqq$$

$$\therefore y - \frac{1}{2}(p+q)x + aqq = 0$$



(ii) If PQ passes through $(0, -a)$ then $(0, -a)$ satisfies the equation of PQ

$$\therefore -a - \frac{1}{2}(p+q) \times 0 + aqq = 0$$

$$-a + aqq = 0$$

$$pq = 1$$

(iii)

SP = PM

By the focus-directrix definition

$$SP = MK + KP \\ = a + ap^2 \\ = a(1+p^2)$$

$$\text{Similarly } SQ = a(1+q^2)$$

$$\therefore \frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a(1+p^2)} + \frac{1}{a(1+q^2)}$$

$$\text{But from (ii) } q = \frac{1}{p}$$

$$\therefore \frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a(1+p^2)} + \frac{1}{a(1+\frac{1}{p^2})}$$

$$= \frac{1}{a(1+p^2)} + \frac{p^2}{a(1+p^2)}$$

$$= \frac{1+p^2}{a(1+p^2)}$$

$$= \frac{1}{a} \text{ as required}$$

c) let the roots be

$$\alpha, \beta \text{ and } \alpha + \beta$$

$$\therefore \alpha + \beta + (\alpha + \beta) = -\frac{b}{a}$$

$$\text{i.e. } 2\alpha + 2\beta = -\frac{b}{a}$$

$$\alpha + \beta = -\frac{b}{2a}$$

But $\alpha + \beta$ is a root of

$$x^3 + px^2 + qx + r = 0$$

$$\therefore \left(-\frac{b}{2a}\right)^3 + p\left(-\frac{b}{2a}\right)^2 + q\left(-\frac{b}{2a}\right) + r = 0$$

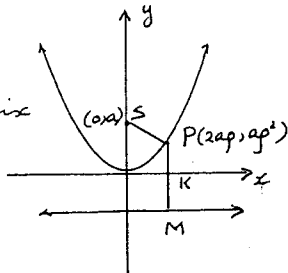
$$-\frac{p^3}{8} + \frac{p^3}{4} - \frac{pq}{2} + r = 0$$

$$-p^3 + 2p^3 - 4pq + 8r = 0$$

$$\underline{p^3 + 8r = 4pq}$$

Question 6.

$$a) \cos x - \sqrt{3} \sin x = 1$$



$$\frac{1-t^2}{1+t^2} - \sqrt{3} \frac{2t}{1+t^2} = 1$$

$$\text{where } t = \tan \frac{x}{2}$$

$$\therefore 1-t^2 - 2\sqrt{3}t = 1+t^2$$

$$-2\sqrt{3}t = 2t^2$$

$$0 = 2t^2 + 2\sqrt{3}t$$

$$0 = 2t(t + \sqrt{3})$$

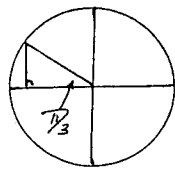
$$2t = 0 \text{ or } t + \sqrt{3} = 0$$

$$x = 0 \quad x = -\sqrt{3}$$

$$\tan \frac{x}{2} = 0 \text{ or } \tan \frac{x}{2} = -\sqrt{3}$$

$$\text{where } 0 \leq \frac{x}{2} \leq \pi$$

$$\frac{x}{2} = 0, \pi \\ x = 0, 2\pi$$



$$\frac{x}{2} = \frac{2\pi}{3}$$

$$x = \frac{4\pi}{3}$$

$$\therefore x = 0, 2\pi, \frac{4\pi}{3}$$

$$b) (i) V = \frac{1}{3} \pi r^2 h$$

$$\text{But } r = \frac{1}{2} h$$

$$\therefore V = \frac{1}{3} \pi \times \left(\frac{1}{2} h\right)^2 h$$

$$= \frac{1}{3} \pi \times \frac{1}{4} h^2 \cdot h$$

$$= \frac{\pi}{12} h^3$$

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$= \frac{\pi}{12} \times 3h^2 \frac{dh}{dt}$$

$$\text{Now } \frac{dV}{dt} = 20 \text{ cm}^3/\text{s}$$

$$\therefore 20 = \frac{\pi}{12} \times 3h^2 \frac{dh}{dt}$$

$$\underline{\underline{\frac{dh}{dt} = \frac{80}{\pi h^2}}}$$

$$(ii) \frac{dh}{dt} = \frac{80}{\pi h^2}$$

$$\text{If } h = 8, \frac{dh}{dt} = \frac{80}{\pi \times 64} \\ = \frac{5}{4\pi} \text{ cm/sec}$$

$$(iii) \frac{dh}{dt} = \frac{80}{\pi h^2}$$

$$\therefore \frac{dt}{dh} = \frac{\pi h^2}{80}$$

$$t = \frac{\pi}{80} \int h^2 dh$$

$$t = \frac{\pi}{80} \frac{h^3}{3} + C$$

$$\text{at } t = 0, h = 0$$

$$\therefore 0 = 0 + C \\ C = 0$$

$$t = \frac{\pi h^3}{240}$$

$$\text{when } h = 8$$

$$t = \frac{\pi \times 8^3}{240}$$

$$= \frac{32\pi}{15} \text{ sec}$$

$$\approx 6.7 \text{ sec.}$$

c)

$$\text{Let } AB = 2a \text{ units}$$

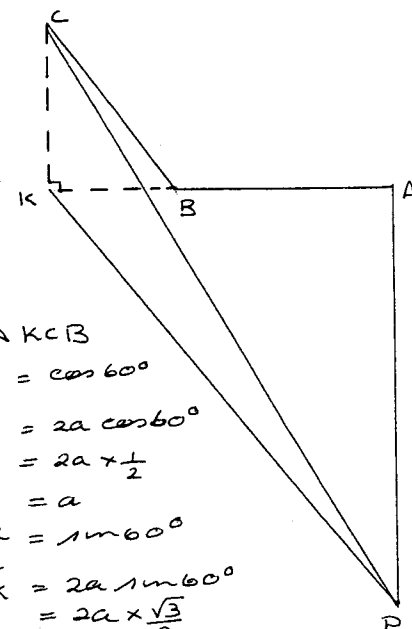
$$\therefore AP = 4a \text{ units}$$

Each $\angle$ of a regular

hexagon

$$= \frac{(6-2) \times 180^\circ}{6}$$

$$= 120^\circ$$

In $\triangle KCB$

$$\frac{KB}{2a} = \cos 60^\circ$$

$$KB = 2a \cos 60^\circ$$

$$= 2a \times \frac{1}{2}$$

$$= a$$

$$\frac{CK}{2a} = \sin 60^\circ$$

$$CK = 2a \sin 60^\circ$$

$$= 2a \times \frac{\sqrt{3}}{2}$$

$$= \sqrt{3}a$$

In $\triangle KAP$

$$(KP)^2 = (3a)^2 + (4a)^2 \text{ Pythagoras'}$$

$$(KP)^2 = 9a^2 + 16a^2 \text{ theorem}$$

$$= 25a^2$$

$$\therefore KP = 5a$$

$$\therefore \tan \angle CPK = \frac{\sqrt{3}a}{5a} \\ = \frac{\sqrt{3}}{5}$$

Question 7

$$f(x) = x^2 - 6x + 13$$

$$y = x^2 - 6x + 13$$

$$y = x^2 - 6x + 9 + 13 - 9$$

$$y = (x-3)^2 + 4$$

$$y - 4 = (x-3)^2$$

$x \geq 3$ is the largest domain for which $f(x)$ has an inverse

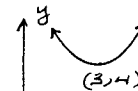
$$(ii) (y-4) = (x-3)^2$$

$$\therefore \text{Inverse is}$$

$$(x-3)^2 = (y-4)$$

$$y-3 = \pm \sqrt{x-3}$$

$$y = 3 \pm \sqrt{x-3}$$



Note for $f(x)$

Domain $x \geq 3$

Range $y \geq 4$

$\therefore$ for $f^{-1}(x)$

Domain $x \geq 4$

Range $y \geq 3$

$$\therefore \underline{f^{-1}(x) = 3 + \sqrt{x-4}}$$

$$c(i) (1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

(ii) Let $n=10$ and $x=7$

$$\therefore (1+7)^{10} = {}^{10}C_0 + {}^{10}C_1 7 + {}^{10}C_2 7^2 + \dots + {}^{10}C_{10} 7^{10}$$

Let $n=10$, $x=-7$

$$\therefore (1+(-7))^{10} = {}^{10}C_0 - {}^{10}C_1 7 + {}^{10}C_2 7^2 - {}^{10}C_3 7^3 + \dots - {}^{10}C_9 7^9 + {}^{10}C_{10} 7^{10}$$

Add

$$\therefore (1+7)^{10} + (1-7)^{10} = 2 {}^{10}C_0 + 2 {}^{10}C_2 7^2 + \dots + 2 {}^{10}C_{10} 7^{10}$$

$$8^{10} + (-6)^{10} = 2 \{ {}^{10}C_0 + {}^{10}C_2 7^2 + \dots + {}^{10}C_{10} 7^{10} \}$$

$$(2^3)^{10} + (2 \times 3)^{10} = 2 \{ {}^{10}C_0 + {}^{10}C_2 7^2 + \dots + {}^{10}C_{10} 7^{10} \}$$

$$\frac{2^{30} + 2^{10} \times 3^{10}}{2} = {}^{10}C_0 + {}^{10}C_2 7^2 + \dots + {}^{10}C_{10} 7^{10}$$

$$\frac{2^{10}(2^{20} + 3^{10})}{2} = {}^{10}C_0 + {}^{10}C_2 7^2 + \dots + {}^{10}C_{10} 7^{10}$$

$$\therefore {}^{10}C_0 + {}^{10}C_2 7^2 + {}^{10}C_4 7^4 + {}^{10}C_6 7^6 + {}^{10}C_8 7^8 + {}^{10}C_{10} 7^{10}$$

$$= \underline{2^9(2^{20} + 3^{10})}$$

c (i)

at $t=0$, $x=0$, $y=0$, $\dot{x} = v \cos \alpha$, $\dot{y} = v \sin \alpha$

$$\ddot{x} = 0$$

$$\dot{x} = \int 0 dt$$

$$\dot{x} = C_1$$

$$\text{at } t=0, \dot{x} = v \cos \alpha$$

$$v \cos \alpha = C_1$$

$$\dot{x} = v \cos \alpha$$

$$x = \int (v \cos \alpha) dt$$

$$x = vt \cos \alpha + C_2$$

$$\text{at } t=0, x=0$$

$$\therefore 0 = 0 + C_2$$

$$0 = C_2$$

$$\underline{x = vt \cos \alpha}$$

$$\ddot{y} = -g$$

$$\dot{y} = \int -g dt$$

$$\dot{y} = -gt + C_3$$

$$\text{at } t=0, \dot{y} = v \sin \alpha$$

$$v \sin \alpha = 0 + C_3$$

$$v \sin \alpha = C_3$$

$$\dot{y} = -gt + v \sin \alpha$$

$$y = \int (-gt + v \sin \alpha) dt$$

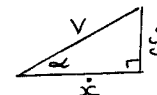
$$y = -\frac{g}{2} t^2 + vt \sin \alpha + C_4$$

$$\text{at } t=0, y=0$$

$$\therefore 0 = 0 + 0 + C_4$$

$$0 = C_4$$

$$\underline{y = -\frac{g}{2} t^2 + vt \sin \alpha}$$



(ii) PUMP 1

$$v = 30$$

$$\angle \text{ of firing} = \alpha$$

$$\ddot{x} = 0$$

$$\dot{x} = 30 \cos \alpha$$

$$x = 30t \cos \alpha$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + 30 \sin \alpha$$

$$y = -5t^2 + 30t \sin \alpha$$

$$y = -5t^2 + 30t \sin \alpha$$

Maximum range occurs

$$\text{when } \alpha = 45^\circ$$

$$\therefore x = 30t \cos 45^\circ, y = -5t^2 + 30t \sin 45^\circ$$

$$x = 30t \times \frac{1}{\sqrt{2}} = -5t^2 + 30t \times \frac{1}{\sqrt{2}}$$

$$\text{when } y=0, -5t^2 + \frac{30t}{\sqrt{2}} = 0$$

$$-5t(t - \frac{6}{\sqrt{2}}) = 0$$

$$\therefore t=0 \text{ or } t = \frac{6}{\sqrt{2}}$$

$$= 3\sqrt{2}$$

$$\text{when } t = 3\sqrt{2}, x = 30 \times 3\sqrt{2} \times \frac{1}{\sqrt{2}}$$

$$= 90$$

$\therefore$ Pump A will reach the fire

PUMP 2

$$\alpha = 0$$

$$v = 40$$

$$\therefore x = 40t \cos 0, y = -5t^2 + 40t \sin 0$$

$$x = 40t \quad y = -5t^2$$

$$\text{when } y = -5$$

$$-5 = -5t^2$$

$$\therefore t^2 = 1$$

$$t = 1 \quad \text{Note } t > 0$$

$$\text{If } t=1, x = 40 \times 1$$

$$x = 40$$

$\therefore$ Pump 2 does not

reach the fire

* It has to reach 60 metres to reach the fire